

Team Technical Documentation

Team: MIRU1

Institution: Hanyang University

Vehicle Hardware

- Chassis Platform: Traxxas Slash 4×4
- Battery: 3S LiPo × 1
- Motor: Traxxas Velineon 3500 Brushless Motor
- ESC: VESC 6 MK VI
- Main Computer: NVIDIA Jetson Orin Nano
- LiDAR: Hokuyo UST-10LX

Software Stack

- OS: Ubuntu 22.04
- Middleware: ROS2 Humble
- Programming Languages: C++, Python
- Localization Algorithm: Particle Filter
- Control Algorithm: MAP / MPC
- Detection and Avoidance Algorithm: Lane Switching

Detection and Avoidance

The vehicle uses a lane switching algorithm based on LiDAR data to avoid obstacles and safely navigate the track.

Team Vehicle

